

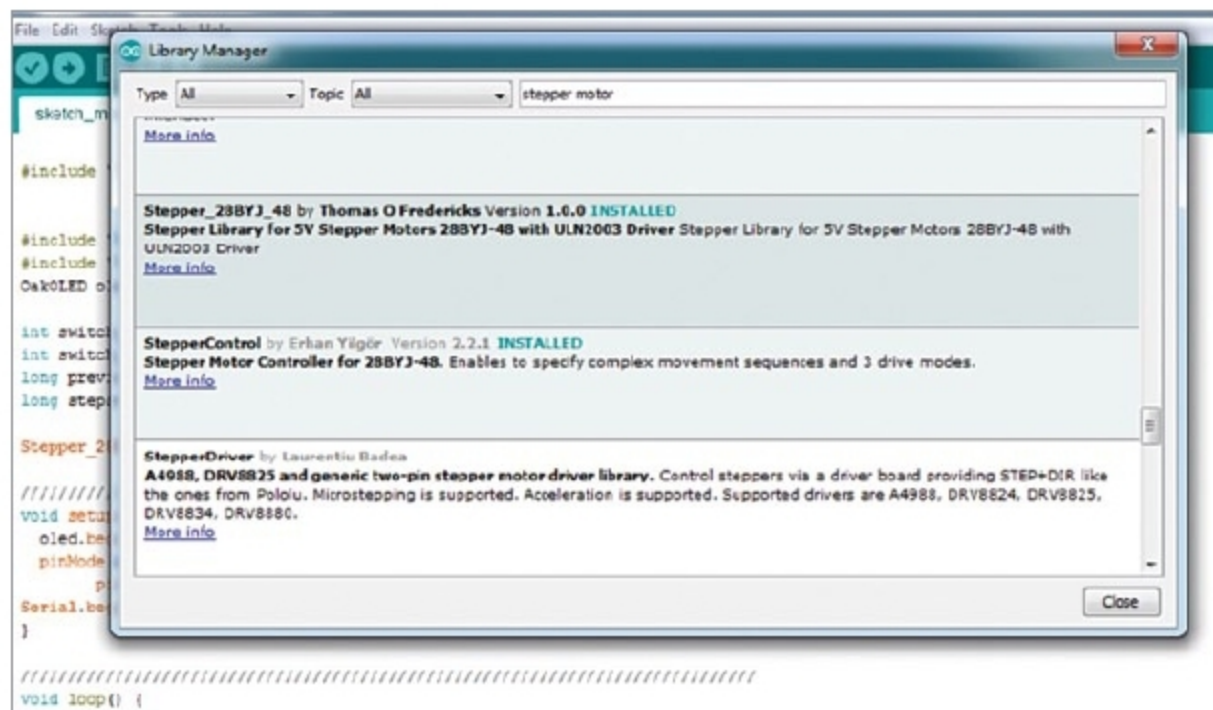


Fig. 3: Installing stepper motor libraries

```
sketch_mar19a$

#include "Stepper_28BYJ_48.h"

#include "Wire.h"
#include "OakOLED.h"
OakOLED oled;

int switch_1_pin = 13;
int switch_2_pin = 12;
long previous;
long steps=0;

Stepper_28BYJ_48 stepper(8,9,10,11);

void setup() {
  oled.begin();
  pinMode(switch_1_pin, INPUT_PULLUP);
  pinMode(switch_2_pin, INPUT_PULLUP);
  Serial.begin(9600);
}

void loop() {
```

Fig. 4: Arduino code

```
void setup() {
  oled.begin();
  pinMode(switch_1_pin, INPUT_PULLUP);
  pinMode(switch_2_pin, INPUT_PULLUP);
  Serial.begin(9600);
}

void loop() {
```

Fig. 5: Setup function

```
void loop() {

  if ( digitalRead(switch_1_pin) == LOW ) {
    stepper.step(-1);
    steps= previous - 1;
    previous=steps;
  } if ( digitalRead(switch_2_pin) == LOW ) {
    stepper.step(1);
    steps= previous + 1;
    previous=steps;
  }
  int angle =map(steps,0,510,0,360);
  Serial.println(angle);

  oled.clearDisplay();
  oled.setTextSize(1);
  oled.setTextColor(1);
  oled.setCursor(0, 0);
  oled.println("Angle");
  oled.setTextSize(2);
  oled.setCursor(10, 12);
  oled.println(angle);
  oled.setTextSize(2);
  oled.setCursor(80, 12);
  oled.println("Degree");
  oled.display();
}
```

Fig. 6: Arduino code for angle calculation

installation of the required libraries, start Arduino coding.

Include stepper motor libraries and declare the variables needed to store different values as shown in Fig. 4. After this create a setup () function, which is shown in Fig. 5.

Next, create a loop function and include 'if () condition' to check the state of the pushbutton. When if () condition in code says true, stepper motor moves one step and stays there until the condition turns false (which happens when the switch is not pressed).

To measure the angle, calculate the steps moved by the motor. To do so, I have created a step variable that will store the previous value of stepper motor and go on adding

one by one as the motor moves. Are you confused? Let me explain it in simple words.

Suppose the initial value of the step variable is zero. When the motor moves one step, its value will change to 0 + 1. This value is assigned to another variable named 'previous,' so the previous value is now 1; when the motor moves another step, the new value becomes previous value + 1. In this way, each time the motor moves a step, the number of steps automatically get added. For example,

Step = 0 + 1

Previous = 0

Step = 1 + 1

Previous = 1

Step = 2 + 1

Previous = 2

This is how the steps of motor are calculated.

After this, we need to convert the steps of motor into angle by using a map () function.

```
map (min steps, max step, min angle,
max angle)
```

The magic happens here! When we put this code in loop function, the steps automatically get converted into angle.

In order to map the angle value of stepper motor, we need to know the maximum number of steps it takes to complete one full rotation. In this project, stepper motor takes 510 steps to complete one full rotation. The same is defined in the code as shown in Fig. 6.

Now we are done with the coding part. So let's construct the digital pro-



Fig. 7: Digital protractor